

C++ Cheat Sheet for Java Programmers

Your Name

Department / University

1. Basic Syntax

- Include headers: `#include <iostream>, #include <cmath>, #include <string>`
- Entry point: `int main() { ... }`
- Semicolons end all statements.
- Comments: `//` (single line), `/* ... */` (multi-line)

2. Data Types

- Integers: `int`, `long`, `long long`
- Unsigned: `unsigned int`, `unsigned long`
- Floating point: `float`, `double`
- Boolean: `bool`
- Character: `char`
- String: `std::string`
- References: `int& r = x;` (alias to variable)

3. Variables and Constants

- Declaration: `int age;`
- Initialization: `double pi = 3.14159;`
- Constants: `const int MAX_VALUE = 100;`

4. Input and Output

- Input: `cin >> x;`
- Output: `cout << "Hello" << endl;`
- Fixed precision: `cout << fixed << setprecision(2) << value;`

5. Operators

- Arithmetic: `+`, `-`, `*`, `/`, `%`
- Comparison: `==`, `!=`, `>`, `<`, `>=`, `<=`
- Logical: `&&`, `||`, `!`
- Bitwise: `&`, `|`, `^`, `~`, `<<`, `>>`
- Assignment: `+=`, `-=`, `*=`, `/=`, `%=`
- Increment / decrement: `++`, `--`

6. Control Flow

- If/else: `if(cond){...} else {...}`
- Switch: `switch(x){ case 1: ...; break; }`
- Loops:
 - `for(init; cond; inc){...}`
 - `while(cond){...}`
 - `do{...} while(cond);`
- `break`, `continue`

7. Functions

- Syntax: `return_type name(params){...}`
- Overloading: same name, different parameters.
- Default parameters: `int f(int x=0);`

8. Arrays

- Declaration: `int a[10];`
- Access: `a[0]`

9. Pointers

- Declare: `int* p;`
- Address-of: `&x`
- Dereference: `*p`

10. Classes and Objects

- Class = data + methods
- Create objects: `MyClass obj;`

11. Strings

- Include: `#include <string>`
- Create: `string s = "Hello";`
- Methods: `length()`, `substr()`, `find()`

12. Standard Template Library (STL)

- Containers: `vector`, `list`, `map`, `set`
- Algorithms: `sort()`, `find()`
- Iterators for traversal

13. Memory Management

- Dynamic allocation: `int* p = new int;` to free memory `delete p;`
- Allocate an array: `int *p=new int[20];` to free memory `delete p[];`
- Free memory: `delete p;`
- Smart pointers: `unique_ptr`, `shared_ptr`

14. File Input/Output

- `ifstream` → read files
- `ofstream` → write files
- `fstream` → both

15. Namespaces

- Avoid conflicts using namespaces.
- `using namespace std;`
- Or fully qualify: `std::cout`

16. Exception Handling

- `try {...} catch(exception& e) {...}`
- `throw` to signal errors

17. OOP Concepts

- Encapsulation, inheritance, polymorphism
- Access: `public`, `private`, `protected`
- Constructors / destructors

18. Templates

- Write generic code:
`template <typename T> T max(T a, T b){...}`

Logical Expression Evaluation in C++

Logical Operators:

- AND: `&&`
- OR: `||`
- NOT: `!`

Short circuit:

- `false && expr` → right side not evaluated
- `true || expr` → right side not evaluated

Key Differences from Java

- No garbage collection (manual memory management)
- Direct pointer manipulation
- Multiple inheritance allowed
- Operator overloading
- Lower-level control of hardware

C++ Hello World

```
#include <iostream>
using namespace std;

int main() {
    cout << "Hello_World" << endl;
    return 0;
}
```